

IMMUNOLOGY

Origin of immunoglobulins and T cell receptors: A candidate gene for invasion by the RAG transposon

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Rearranging antigen receptors (AgRs) arose when a variable (V) domain exon was invaded by the recombination-activating gene (RAG) transposon ~500 million years ago. We show here that the elasmobranch immunoglobulin heavy (IgH) isotypes—IgM, IgW, and IgNAR—are linked near the $\alpha\delta$ T cell receptor (TCR $\alpha\delta$) locus. This linkage presages the emergence of the osteichthyan IgH translocon arrangement and clarifies the relationship between IgH and TCR δ s. Recently, we reported *Urlg*, a nonrearranging, elasmobranch major histocompatibility complex (MHC)-linked AgR gene. Here, we describe a nonrearranging *Urlg* paralogue, *Urlg2*, linked to this IgM/IgNAR/IgW/TCR $\alpha\delta$ gene cluster in an AgR complex (AgRC). *Urlg2* amino-terminal domains make homodimers where the C2-C3 structure resembles IgGfC. A relative of the *Urlg2* V domain exon was invaded by the RAG transposon, revealing the genesis of the adaptive immune system. Our data indicate that an ancestral chromosome encoded an AgR precursor, undergoing RAG-mediated rearrangement after genome-wide duplication on one chromosome and retaining nonrearranging relics in the MHC and AgRC.

INTRODUCTION

In jawed vertebrates (gnathostomes), antigen receptors (AgRs), immunoglobulins (Igs), and T cell receptors (TCRs), as well as the major histocompatibility complex (MHC) class I and class II proteins, define humanlike adaptive immunity (1–3). Ig and TCR are made up of unique members of the Ig superfamily (IgSF) domains and share similar cardinal structures, with membrane-distal variable (V; or in the case of AgR, VJ) domains that bind antigen, and membrane-proximal constant (C; or in the case of AgR, C1) domains involved in signaling and/or effector functions (4). The V domains are diverse by virtue of their generation by recombination-activating gene (RAG)-mediated somatic gene rearrangements of V to J [for Ig light (IgL) chains and TCR α or TCR γ chains] or V to D to J [for Ig heavy (IgH) chains and TCR β or TCR δ chains] (5, 6). It has been hypothesized that Ig and TCR were generated early in vertebrate history when a nonrearranging precursor VJ gene was invaded by the RAG transposon, which split a single VJ exon into separate V and J gene segments (7, 8). Searching for genes with intact single-exon VJ genes has been of great interest to examine the origins of Igs and TCRs and, thus, the advent of gnathostome adaptive immunity (9–11).

The timing of the appearance of humanlike AgR-based adaptive immunity coincides with the second genome-wide duplication event (referred to as 2R), which occurred just before the emergence of

jawed vertebrates, while the first event (1R) predated the advent of all vertebrates (12, 13). The 2R is associated with a large expansion and neo/subfunctionalization of immune (and other) genes (14, 15). Several lines of evidence suggest that the RAG transposon insertion, as well as the appearance of MHC class I or II and immunoproteasome genes, occurred not long after the 2R (10, 16, 17).

Evolutionarily, AgR-based adaptive immunity appeared when jawed vertebrates (gnathostomes) emerged ~500 million years ago (Ma), with cartilaginous fish (sharks, rays, and chimeras) being the oldest living taxon with AgR and MHC (1, 18). In all gnathostomes, there are multiple isotypes for both IgH and IgL chains. In humans, IgH chain genes are found on chromosome 14 (hchr14) in the translocon organization, containing separate clusters of V, D, and J segments followed by C region genes for all isotypes. Theoretically, any V, D, and J segment can be joined together by RAG-mediated rearrangement to generate a highly diverse repertoire of different V domains. In contrast, all cartilaginous fish IgH and IgL chain genes are arranged in “miniclusters” where each V-(D)-J-C is found as a cassette and rearrangement usually occurs within a cluster, likely the ancestral gnathostome condition (18, 19). The number of these IgH/L miniclusters varies depending on the cartilaginous fish species, and their distribution in the genome is not well known (19, 20).

TCR α/δ and IgH genes are linked on hchr14, a condition often found in other vertebrates, including sharks, frogs, coelacanth, and nonplacental mammals; however, the physical distance between these genes is much closer in these non-eutherian vertebrates (21–25). Transrearrangements of IgM and IgW V regions to TCR δ D segments are common in sharks and some other nonmammalian species (26, 27). Sharks also have a doubly rearranging TCR δ gene, NARTCR, which encodes three Ig domains, where an “extra” rearranging V from IgNAR is perched on the canonical TCR δ VC to make up a three-domain AgR chain (28). The presence of these chimeric Ig-TCR molecules in sharks suggests that their close proximity by linkage allows interlocus recombination, both ongoing within an individual and over evolutionary time to form novel AgR loci (27). Here, we examined this AgR locus using various cartilaginous fish genomes and uncovered novel genes and gene organization.

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Previous genomic and bioinformatic studies suggested that the AgR, as well as the MHC per se, originated from the proto-MHC, before the appearance of adaptive immunity (10, 16, 17). Recently, we found an MHC-linked Ig gene with a nonsplit (nonrearranged) V domain, *UrIg* (ancient Ig or unrearranged Ig) (29, 30). *UrIg* is a disulfide-linked homodimer, with each chain having a VJ domain followed by three C1-type domains. AlphaFold modeling suggested a notable similarity to mammalian IgG (without IgL chains) by convergent evolution, but *UrIg* only has a secreted form and is produced by the liver, not by lymphocytes. *UrIg* protein was found in various tissues in association with the extracellular matrix (ECM). Phylogenetic and genetic analyses suggested that *UrIg* is related to the precursor of AgR.

In this study, we uncovered the genomic location of the three IgH isotypes in sharks, allowing us to infer the transition in the evolution of Ig loci from a minicuster in cartilaginous fish to translocon in all other gnathostomes. During our search, we serendipitously found a *UrIg* paralogue, *UrIg2*, linked to this complex. Our detection of the nonrearranging *UrIg2* genetically linked to the Ig/TCR genes likely unveils the timing of the rearranging AgR emergence early in vertebrate evolution, before the 2R. This nonrearranging AgR gene is the most promising candidate to be closely related to the VJ gene invaded by the RAG transposon and further reinforces the ancestry of the gnathostome AgR in the proto-MHC.

RESULTS

Close linkage of all IgH isotypes and some IgL loci to the TCR $\alpha\delta$ complex and PSMB5 in the cartilaginous fish genome

Cartilaginous fish have three IgH [IgM, IgW (IgD equivalent), and IgNAR] and four IgL (λ , κ , σ , and σ -2) isotypes, all found in the minicuster organization (19, 31). The number of clusters for any isotype is species dependent, ranging from a single copy to as many as 100 loci. In recent years, several shark genomes have been sequenced and assembled at the chromosomal level. Therefore, we embarked on determining the chromosomal location(s) of these IgH clusters. We used Ig and TCR C domains of nurse shark AgR genes to search for gene-model orthologues in other sharks, as the similarity is generally high between elasmobranch species (>80%) (32). All gene models in the scaffolds were manually examined for the presence of Ig and TCR genes and the constitutive proteasome component *PSMB5*, which maps near IgH/TCR $\alpha\delta$ on huchr14 (discussed below in the context of whole-genome duplications).

Previous work in the holocephalan elephant shark (*Callorhynchus milli*) and elasmobranch nurse shark (*Ginglymostoma cirratum*) showed that some V regions of immunoglobulin heavy chain (V_H) genes were linked to the TCR $\alpha\delta$ locus (25, 33). Now, the best-assembled shark and ray genomes with chromosome-level assemblies were used to confirm and extend the gene linkage studies. Gene arrangements in the large scaffold of four cartilaginous fish species (great white shark, catshark, epaulette shark, and stingray) were compared to human huchr14. We found that shark and ray IgH (and some IgL) genes were closely linked to TCR $\alpha\delta$ in these four elasmobranch species (Fig. 1 and table S1). Unexpectedly, all three shark IgH isotypes were found in this linkage group, syntenic with *PSMB5*. The AgR genes, as expected, were found in the translocon and minicuster organization for TCR $\alpha\delta$ and IgH, respectively. The V(D)JC of each IgH (and IgL) isotype clustered together, not intermingling with other IgH isotypes. However, some V domain exons

of the TCR $\alpha\delta$ type were found between the IgH gene clusters, and other V domains were intermediate in sequence between those of TCR and Ig.

Not all cartilaginous fish Ig genes are present in this newly found AgR complex (AgRC) (34), but the fact that all three IgH isotypes, and some IgL, are linked strongly suggests that they all arose in this region, consistent with previous analyses of their phylogenetic relationships (35, 36). Transrearrangements from IgM/W V_H segments to TCR δ D segments are also easily supported by their close proximity (27). The presence of the single-domain V in both IgNAR and NARTCR is also clear: There was a transfer from one of the linked loci to the other, most likely the NARTCR V to an IgW cluster (Fig. 2) (28, 37). Thus, we hypothesize that the linkage of the three elasmobranch IgH clusters provided the gist for the streamlined translocon organization in all higher vertebrates (38). IgM and IgW (IgD) are primordial isotypes found throughout the vertebrate lineage (39, 40), and the two isotypes are shared by cartilaginous and bony fish; thus, it is likely these two genes established the foundation of the translocon configuration (Fig. 2).

UrIg2, the MHC-linked *UrIg* paralogue, found in the AgRC

During the analysis of this AgRC, we noticed a gene that was “automatically annotated” as an IgL chain, but it was not any of the four elasmobranch IgL genes, as there were multiple C-encoding exons. Instead, this gene frequently hit *UrIg*, an MHC-linked *UrIg*/TCR-like gene that we recently reported (29). This genetic localization and phylogenetic analysis suggested that this could be an ancient paralogue of *UrIg*, and it was named *UrIg2* (Figs. 1, 3, 4, and table S1). Similar to *UrIg*, *UrIg2* has a typical VJ domain encoded by one exon and the canonical “split leader” found in all AgR genes (fig. S1B) (41). Some gene models from the genome sequencing were incomplete; thus, we searched for the entire sequence in RNA sequencing (RNA-seq) databases and found a partial sequence in the liver database (GIWU01221083). By using this partial sequence, we then found a full-length *UrIg2* in a publicly available nurse shark gill RNA-seq database. Nurse shark *UrIg2* contains a VJ and nine C1-type domains and a transmembrane (TM)/cytoplasmic domain (Fig. 3 and fig. S1A). From the phylogenetic tree, the N-terminal three C domains are unique and paralogues of the *UrIg* N-terminal C domains (Fig. 4C), but the other six domains are highly conserved duplicates of domains C2 and C3 (Figs. 3 and 4C and figs. S1A and S2C). These repeated sequences are nearly identical and were likely generated by three gene duplications of two adjacent exons and homogenized by gene conversion (Figs. 3 and 4C and fig. S1A; discussed later). Gene duplication of two adjacent exons was previously shown for IgW and IgNAR from a four C domain precursor (Fig. 2) (35, 36).

The huchr14 and its equivalent in other vertebrates have been hypothesized to be derived from an MHC paralogous region, chr9 in humans (16). The constitutive proteasome subunit *PSMB5* (*LmpX*) gene is found in this complex, while the immunoproteasome interferon-inducible form *PSMB8* (*Lmp7*) is encoded in the MHC on huchr6. These findings suggested to several investigators that AgRs arose in the proto-MHC, an idea fortified by the discovery of *UrIg* in the shark MHC (29), as well as other VJ single exons [e.g., *NCR3* Fig. 4A (42)] in the MHC and MHC paralogous regions (10, 11, 43). Because the paralogue *UrIg2* is encoded in the AgRC with TCR and bona fide IgH/L genes, this finding reinforces (and virtually proves) the hypothesis that AgR arose in the proto-MHC. The most parsimonious interpretation is that an ancestral *UrIg* gene was encoded in

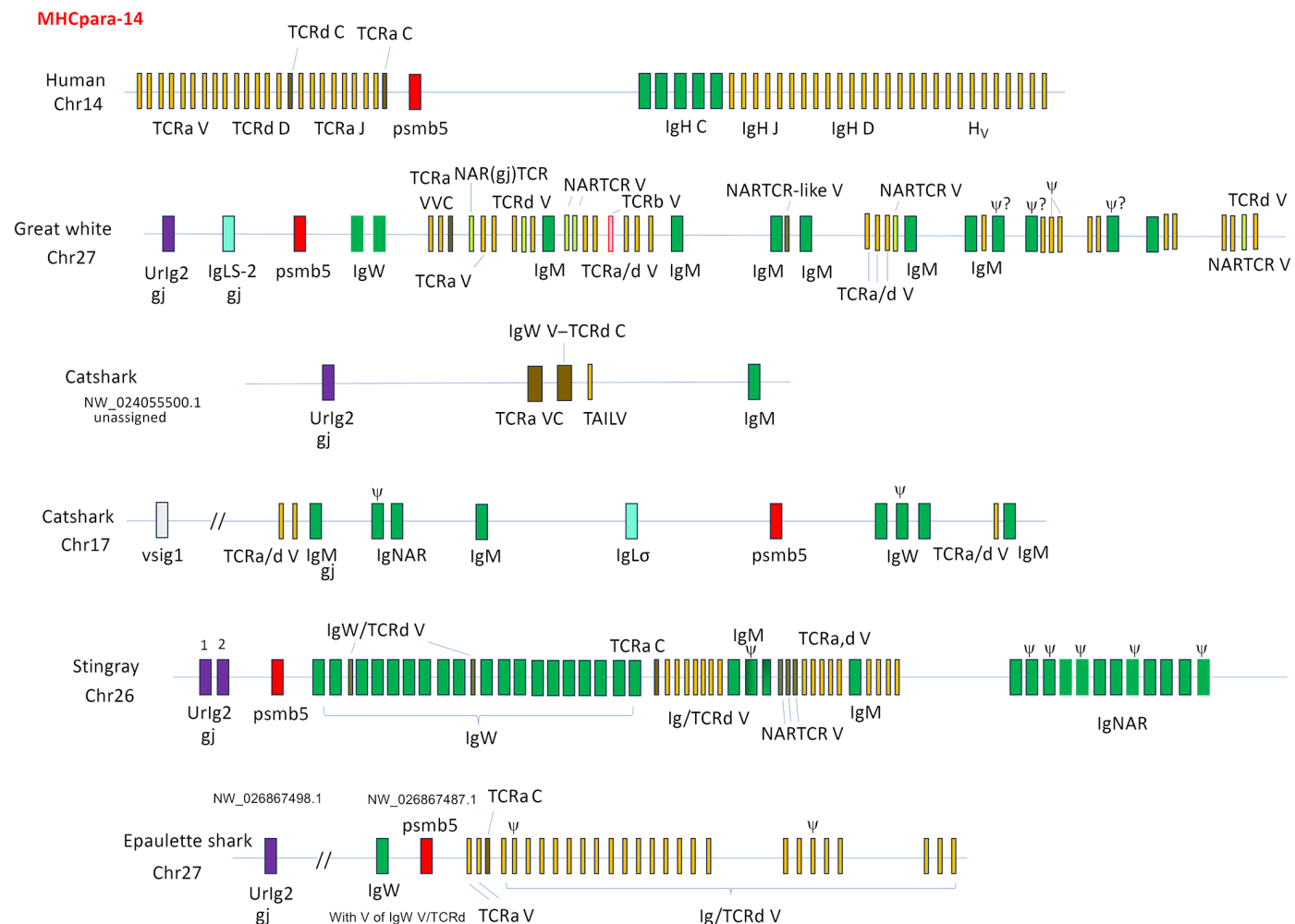


Fig. 1. All IgH chain genes are closely linked together along with *Urlg2*, *TCRαδ*, and some IgL genes in cartilaginous fish. Boxes indicate either genes (multiple exons) or exons (e.g., V domain), and the color is based on the type: All V fragments (orange) except NARTCR V (yellow), genes with C domains of IgH chain (green) and TCR (brown), IgL chain (light blue), and *Urlg2* (purple). *Psm5* (red) is a marker gene, which maps to hchr14 near IgH and TCRαδ loci. The same color theme is used for human genes and gene fragments. All human V fragments (V, D, and J) are shown in orange boxes. The location of genes is not scaled, but coordinates in the scaffolds are listed in table S1. TCRα and δ are shown as TCR a and d.

the hchr6/9 at 1R, and then the two paralogues were generated in the 2R, *Urlg* on hchr6, and *Urlg2* on hchr9 (later translocated to hchr14) (Fig. 2; discussed below).

As expected, because *Urlg* and *Urlg2* separated ~450 Ma, sequence similarity is low between their VJ domains, ranging from 28 to 32% (Fig. 3 and fig. S2). Nevertheless, these two VJ genes cluster together in phylogenetic trees, although distantly, as expected for genes that diverged so long ago (Fig. 4A). The VJ domains of *Urlg* are more conserved between different elasmobranch species (~80% at the amino acid level) (29) than those of *Urlg2*, which share only 50 to 60% identity between different species (fig. S2B). The branch lengths between *Urlg* V of different species are much shorter than those of *Urlg2* (Fig. 4A). In contrast, the conservation of C domains of both *Urlg* and *Urlg2* is similar (50 to 60%) among elasmobranchs (fig. S2C).

Gene models from the genome assemblies were incomplete in many species. We were rigorous in including only those models with a V domain, followed by C domains. Often, TM exons were not

detected by the gene prediction software; however, by comparison to the nurse shark transcript from the RNA-seq database, we were able to find a complete gene model with TM exons in the epauvette shark genome. This shark gene model contained the canonical split leader, one VJ, and 11 C domains, two domains more than nurse shark *Urlg2* (fig. S1, A and B). The number and types of *Urlg2* C domains between sharks and batoids (skates and rays) varied. Shark C1 domains were lost in batoid species, and all batoids contain “hinge” exons [short exons with many prolines (44)] that are absent in sharks (fig. S1, A and C). Thus, in contrast to *Urlg*, which is evolutionarily conserved in its V sequence and a number of C domains among different elasmobranchs, *Urlg2* is divergent in sequence and highly plastic in its gene structure in different species. These data plainly show that the two paralogues are under different selection pressures.

As mentioned, the sequence analyses revealed that the V and C domains of *Urlg2* are more closely related to *Urlg* than to any of the rearranging AgRs. *Urlg2* V domains are the most divergent compared

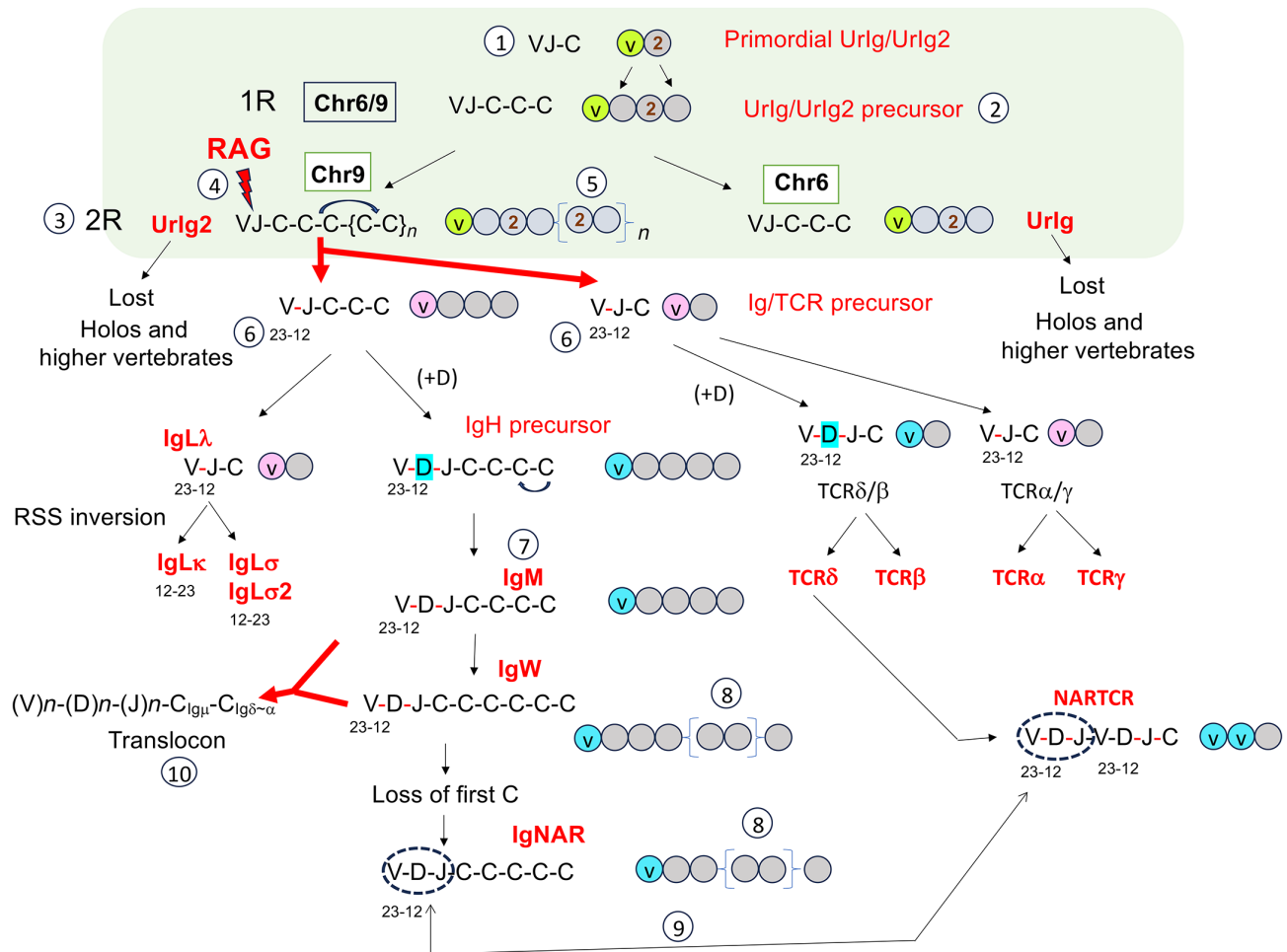


Fig. 2. Speculative origins and descent of the AgR family. Most relevant features here are shaded in green. The *Urlg* precursor is predicted to have been present at 1R on huchr6/9, first as a two-domain gene IgSF VJ (green) and C1 type (gray). The C1-IgSF domain resembles the second C domain of *Urlg/Urlg2* (noted “2” in gray balls) ①. The two domains are likely duplicated to generate a four-domain *Urlg/Urlg2* precursor at 1R in the protoMHC/AgRC ②. After 2R, *Urlg* is found in the MHC (huchr6) and *Urlg2* on huchr9 in the early AgRC ③. We predict that *Urlg2* was duplicated, and its VJ was invaded by the RAG transposon to initiate the rearranging AgR ④ (noted by a dash in V-J); *Urlg2* still exists in elasmobranchs as *Urlg*. There was duplication of the C2/C3 unit in *Urlg2* in sharks (a box ⑤) and further modifications in batoids (hinge formations), while *Urlg* has remained stable in the elasmobranch MHC. The most speculative part of the scheme is the Ig/TCR precursor; the most convincing aspects are that the V23RSS (shown as 23-12) is likely ancestral ⑥, and the *Urlg* precursor “2” domain gave rise to all Ig C domains (see Fig. 4D). In our scheme, IgM is ancestral to IgW ⑦ because of the duplication of the C2/C3 domains in IgW (and IgNAR) (boxed ⑧), but IgM and IgW could have arisen rather from an ancestor with four C domains. The transfer of a NAR V(D)J (dotted circle) from TCRδ to Ig (or vice versa) is shown ⑨. The close linkage of IgM and IgW genes is predicted to later form the IgH translocon organization in higher vertebrates ⑩. V genes with D segments in blue and without D segments in pink.

to the rearranging V domains of clans to which they are genetically linked but, nevertheless, more related to Ig/TCR than *NCR3*, the other nonrearranging, MHC-linked VJ domain gene forming the tree root in Fig. 4B. The *Urlg* and *Urlg2* C domains clustered together based on their positions from N- to C-terminal in the proteins, with the C2 domain being perhaps the oldest member of the family (Figs. 2 and 4C). While it is difficult to determine which *Urlg2* C domains are most similar to particular IgH/L C domains because of the long divergence times and/or the potential for recombinational/gene conversion events at the AgRC, the *Urlg2* C2 domain, again, formed the root of the entire IgC tree (Fig. 4D). The *Urlg2* C1 and C3 domains showed the closest affiliation with each other and with IgL chains, albeit with low bootstrap values (Fig. 4D). There are two other noteworthy features of the phylogenetic analysis: (i) The relationships of the IgM,

IgW, and IgNAR C domains were confirmed (Figs. 2 and 4D) (28, 35, 36), and (ii) the aforementioned gene conversion of the *Urlg2* C domains within a species is convincingly shown for the nurse shark (Gici) and epaulette shark (Heoc) (Fig. 4D).

Urlg was shown to be expressed primarily by the liver, not by lymphocytes, as is found for the rearranging AgR (29, 30). We searched for *Urlg2* expression in various tissues of nurse shark and calculated the TPM (transcripts per kilobase million) read in the RNA-seq (Fig. 5A). The highest level of expression was detected in the gill, followed by the liver and much lower levels in spleen and other tissues. The results of the RNA-seq were further confirmed by semiquantitative reverse transcription polymerase chain reaction (RT-PCR) (Fig. 5B). Thus, both paralogues are not expressed like typical Ig/TCR genes.

A**Leader**

Gici_UrIg2 MTPSVKLLLSLSFCILP DLTSG
 Gici_UrIg .WQVLF..AA.AGPLTA GPLTA

VJ domain

Gici_UrIg2 GVNFRFSSTMTVTGEPEVSMERYSRKTD DRYLLWGYVPRGTS AIKQLLYWNP GS DT
 Gici_UrIg DDTLYI.PISLSRW.**N.T.**TLN.FT..TV.NY..I..SAKP.DLM.SWI...M...SA

Gici_UrIg2 PTYISGKPGRYRAEKNFRQKVMKQIAKATQHDSSRYFCQLTD**NLS****NGNTT**DFCSCGTTLVQA
 Gici_UrIg AL.P.VVSD.FIGS.DWTS.KFS.T..GLR.E..A..Y.GSSNTYFFFS.DWSGCGV.LTVKA.
 C' ' D E F G

C domains

Gici_UrIG2-C1 VPPSSAPLVELHSPFAGQLRE**NGT**ATLLCLASGFYPSNVRISWQMDGRTVTARVSQQPMCQD
 Gici-UrIg-C1 A.SV.P...YFQN.HPEDVFLR**KNIS**...A.ES...DDLSE..RVD.EQ.HRG..HSASLLG

Gici_UrIG2-C1 EREKYTAVSRLQLSSSTWMEAHRYQCTVTHPTLSRPIVKELRTL
 Gici-UrIg-C1 **NGT**...I.K.V**INT**.Q.IKGSII.S.Q.S.GA.PS..I.HISNT

Gici_UrIG2-C2 ASLSLRPTIRMIPPSIREMVTHRTATLGCVATQFIPDPITFSWMAQGLEIPGTSQTLVPTANE
 Gici_UrIG2-C4V..T.....L.....K.....T.....F..M.....
 Gici_UrIG2-C6V.V.....V.....K.....M.....
 Gici_UrIG2-C8QH..V.V.....V.....M.H.....G
 Gici-UrIg-C2 -ERYPME.KLYLLS.TLE.L..MEK...V.....Y.AE.S.H.FIN.NPVVDSEIKIYP.IL.I

Gici_UrIG2-C2 DGSYSARSLLTVALADWESGTSYTCQVPNAQSDPK**NRT**LRVEDA
 Gici_UrIG2-C4E.G.....H.....
 Gici_UrIG2-C6
 Gici_UrIG2-C8G.....
 Gici-UrIg-C2 **N.T**..SQ.E...TSDQM.NR..E...EIQHMSVPDSIIQT**INKTR**

C3 domain

Gici_UrIG2-C3 QLWKQPSLQMTVIE**EEEEDEE**GHVADGGTTLACFVSAFYPGDIYVSWR-EQDTELSNGVTTFPVR
 Gici_UrIG2-C5A.D.E..R.....
 Gici_UrIG2-C7
 Gici_UrIG2-C9T.....
 Gici-UrIg-C3 EFIFLQPITPIKVQSRDSRRPQA.-----G...AG...N..T.R.LQVPGS..N...V

Gici_UrIG2-C3 PDPDGTFTVSVLSVPSRAWHS**GH****NFT**CLVGHQSL-NGLKEAQLRKDS
 Gici_UrIG2-C5
 Gici_UrIG2-C7
 Gici_UrIG2-C9L.....
 Gici-UrIg-C3 R.VS..YS...MTIAFVDVDPTQTYM.E.R.E..TNP.R.RFQIR*

Gici_UrIG2 TM/cyt

GWMDILRKAPWLWGSIAVSLFILCIAVATLVGVHCGRTVSQRKGSYQIPRLDSVPLKEICPSEVDKE*

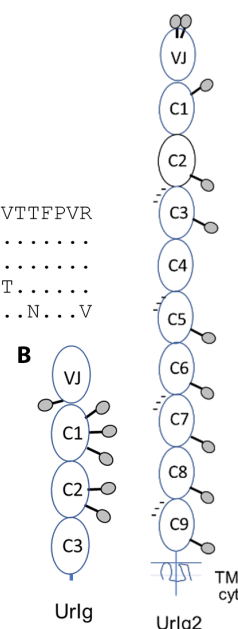


Fig. 3. A domain-by-domain comparison of nurse shark Urlg2 to the paralog Urlg and gene conversion of Urlg2 domains C2 to C9. (A) Deduced amino acid sequences were aligned using BLASTp. The C2, C4, C6, and C8 domains and C3, C5, C7, and C9 domains of Urlg2 are aligned to each other because they are nearly identical. IgSF strand positions in the VJ domain were predicted on the basis of the nurse shark Urlg mature protein sequence, and glycosylation sites are marked with bold letters and underlined. Canonical amino acids typical for immunoglobulin domains are highlighted in gray. Stretches of negatively charged amino acids in the C3, C5, C7, and C9 domains are in bold letters. Dashes and dots indicate absence and identical amino acid sequences compared to nurse shark Urlg, respectively. Asterisks indicate stop codons, and predicted TM residues in Urlg2 are double underlined. The split position of the leader is shown with an arrow. cyt, cytoplasmic. **(B)** The schematic domain structures of Urlg and Urlg2 are shown in the bottom right. Lollipop and - indicate sugars and negatively charged amino acids, respectively.

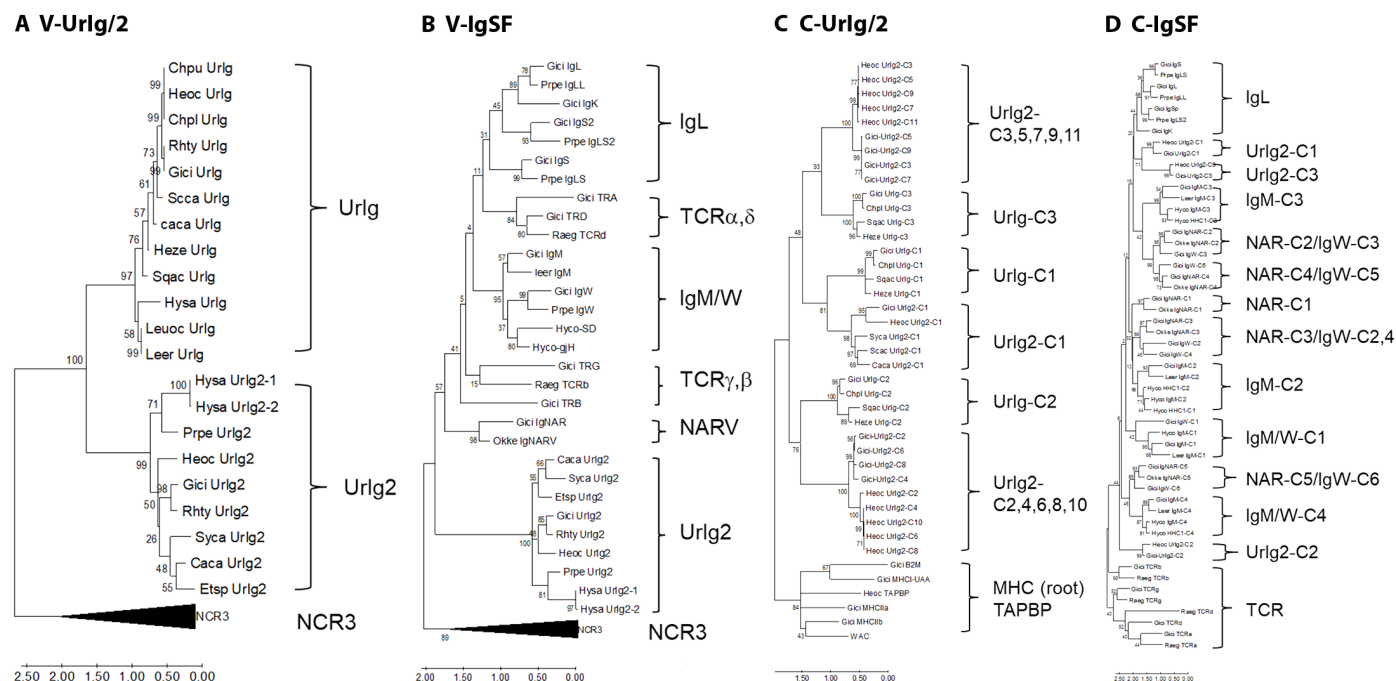


Fig. 4. UrIg2 is most closely related to Urlg but divergent across species. Relationship of UrIg2 domains to AgR domains. Phylogenetic analysis of Urlg and UrIg2 V and C domains [(A) and (C), respectively] and UrIg2 and cartilaginous fish AgR V and C domains [(B) and (D), respectively] are shown. The V (A and B) and C (C and D) IgSF domains are separately compared. Amino acid sequences of IgSF exons were analyzed using bootstrapping (500 runs), maximum likelihood, and Whelan and Goldman model. Numbers on the branch nodes indicate the bootstrap percentage. The tree is drawn to scale with branch lengths measured in the number of substitutions per site, which is shown at the bottom of the trees. (C) is rooted with MHC and tapasin (TAPBP) C1 domains. All other trees are not rooted.

Recombinant UrIg2 V-C1-C2-C3, a disulfide-linked dimer

The four N-terminal V-C1-C2-C3 domains of UrIg2 were expressed in mammalian cells. On the basis of size exclusion chromatography, as well as reducing and nonreducing SDS–polyacrylamide gel electrophoresis, we found the protein to be a disulfide-linked homodimer in solution, with an estimated molecular mass for the glycosylated homodimer of ~125 kDa (expi293Gnt1⁺ cells) or 140 kDa (expi293F cells) (Fig. 6, A and B). Urlg has also been found to form homodimers both recombinantly and in vivo (29).

UrIg2 V-C1-C2-C3 crystal structure

A 2.8-Å-resolution crystal structure was obtained for expi293F-produced V-C1-C2-C3 (Fig. 6D and table S2). The protein forms a ~155-Å-long, head-to-head homodimer in the crystal asymmetric unit. Each IgSF domain contains the canonical disulfide bond linking the B and F strands and the highly conserved Trp “pin” residue (4, 45) in the C strand (figs. S3 and S4) that packs against the canonical disulfide. In addition, two noncanonical disulfide bonds link the V-V domains at residues A112-B112 and A114-B114, as well as a third disulfide bond linking the C1-C1 domains at residues A182-B182 (figs. S3 and S4).

Individual Ig domain structure

The Dali server (46) was used to search for structural homologs for individual V, C1, C2, and C3 domains. The top 20 homologs to each UrIg2 domain are listed in table S3. The V domains are most similar to V_H domains, and C1 domains are most similar to Ig C_κ domains, while C2 domains have similarity with TCR and NKT TCR C_β, TCR C_γ, IgM C_μ4, IgNAR C3, chicken IgY C_γ3, IgG1 C_H1 and C_H2 and IgA C_α3 domains. C3 domains have similarities to IgG1 C_γ3, IgA

C_α3, IgM C_μ4, IgY C_γ4, and TCR C_β and C_γ domains (Fig. 7). There are slightly unusual edge strand positions in the V (A' strand) and C2 (D strand) domains (Fig. 7).

UrIg2 domain pairing

The V domains dimerize with 1117 Å² and 1128 Å² of molecular surface buried on each chain, but the dimer is quite different from that seen in an antibody heterodimer of the variable region of the light and heavy chain (Fv) region (fig. S4, A and B). The UrIg2 V domain dimer interchain contacts are largely between the A and G strands, with the two noncanonical disulfide bonds connecting the G strands (fig. S4A). In contrast, a typical Fv has interchain contacts between opposing β sheet strands GFCC' from each chain. The C1 domains dimerize using the same interface as Fab C_H1 domains, with 654 Å²/674 Å² buried on each domain within the homodimer; however, the pairing angle between the domains is more acute, differing by about 44° from that seen in a typical Fab (fig. S4, C and D), and the distance between the N-terminal ends of each C1 monomer is shorter (28 Å) than, for example, the distance between the N termini of C_κ-C_H1 (~45 Å). The C2-C3 domains dimerize in a manner very similar to that of a C_H2-C_H3 Fc fragment, with almost no interaction between the C2 domains, and the C3 domains forming a C_H3-C_H3-like dimer (fig. S4, E and F). Similar to IgM Fc regions (fig. S4F), there is no carbohydrate filling the gap between C2-C2 in UrIg2; however, Urlg does have an N-linked glycosylation site that we predict would result in carbohydrate occupying this space, analogous to that seen, for example, in an IgG1 Fc (29).

While UrIg2 V formed the root of the AgR tree (Fig. 4B), structurally, it is most similar to V_H (tables S3 and S4). Consistent with

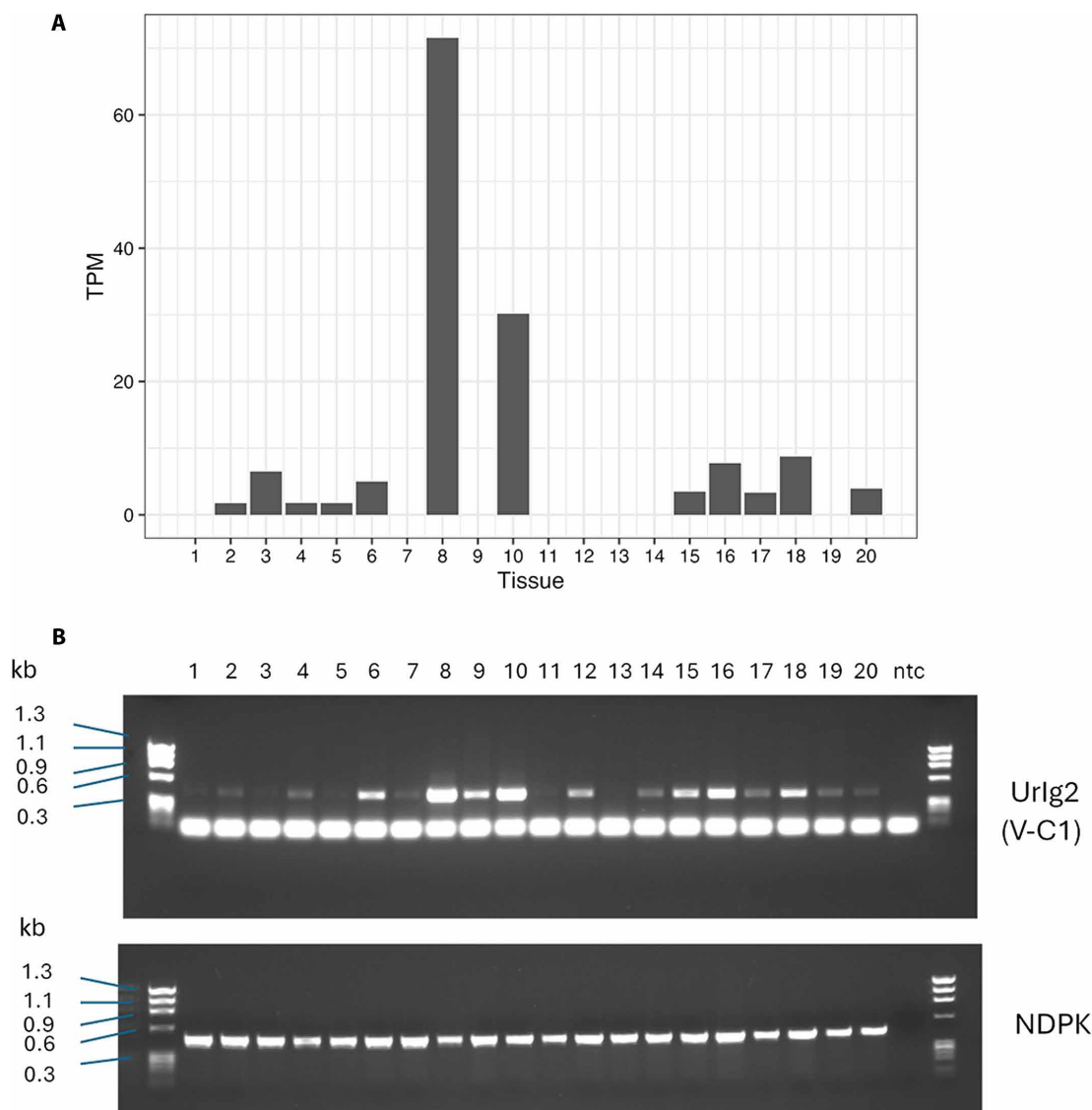


Fig. 5. *UrIg2* is preferentially expressed in shark liver and gill. (A) Bulk RNA-seq from shark NJ. Expression level was calculated using RSEM (see Materials and Methods). Sample: 1-Brain/diencephalon; 2- Brain/cerebellum; 3- Brain/medulla; 4- Brain/midbrain; 5- Brain/cerebrum; 6- colon; 7- epigonal; 8- gill; 9- kidney; 10- liver; 11- muscle; 12- olfactory epithelium; 13- pancreas; 14- rectal gland; 15- spiral valve; 16- spleen; 17- stomach; 18- thymus; 19- white blood cells; 20- uterus. (B) reverse transcription polymerase chain reaction (RT-PCR) from shark NJ RNA. An equal amount of total RNA was used to synthesize cDNA, and an equal amount of cDNA was used for each PCR reaction. Nucleoside-diphosphate kinase (NDPK) is a PCR control. Band sizes (in kilobase) from the Φ X174 DNA/HaeIII Marker are shown on the right and left sides of the agarose gel. Samples same as in (A). ntc, nontemplate control.

the phylogenetic analysis (Fig. 4D), the *UrIg2* C1 domain is structurally most similar to IgL chain C domains. Also consistent with the phylogenetic tree in which the *UrIg2* C2 domain seems most primitive (Fig. 4D), it is not related structurally to a single type of C domain but instead has similarities to many different types (table S3).

DISCUSSION

Genesis of the gnathostome adaptive immune system by the RAG transposon and the *UrIg* family

Immunologists have been keenly interested in the emergence of the adaptive immune system ever since Tonegawa found recombination signal sequences (RSS) in 1979 and predicted that there could have

been a transposon invasion of an intact IgSF gene (7). The discovery of RAG (47) and, later, RAG transposons themselves (48, 49) strengthened this theory, and now, intact transposons have been detected in early chordates and partial transposons throughout the animal kingdom (50). A homolog of *UrIg2* (if not a duplicate of *UrIg2* itself) is the most likely candidate for invasion of the RAG transposon. This innovative event, which permits fortuitous differential sizing of the CDR3 loop (51), would have been expected to initiate a great expansion and divergence of the rearranging genes, including Ig and TCR, initially at the same genetic complex. Of course, one can never prove that a *UrIg*/*UrIg2* homolog was invaded by the RAG transposon, as it occurred 500 Ma, but the gene organization, synteny with both the MHC and AgRC, and structure of the molecules mark

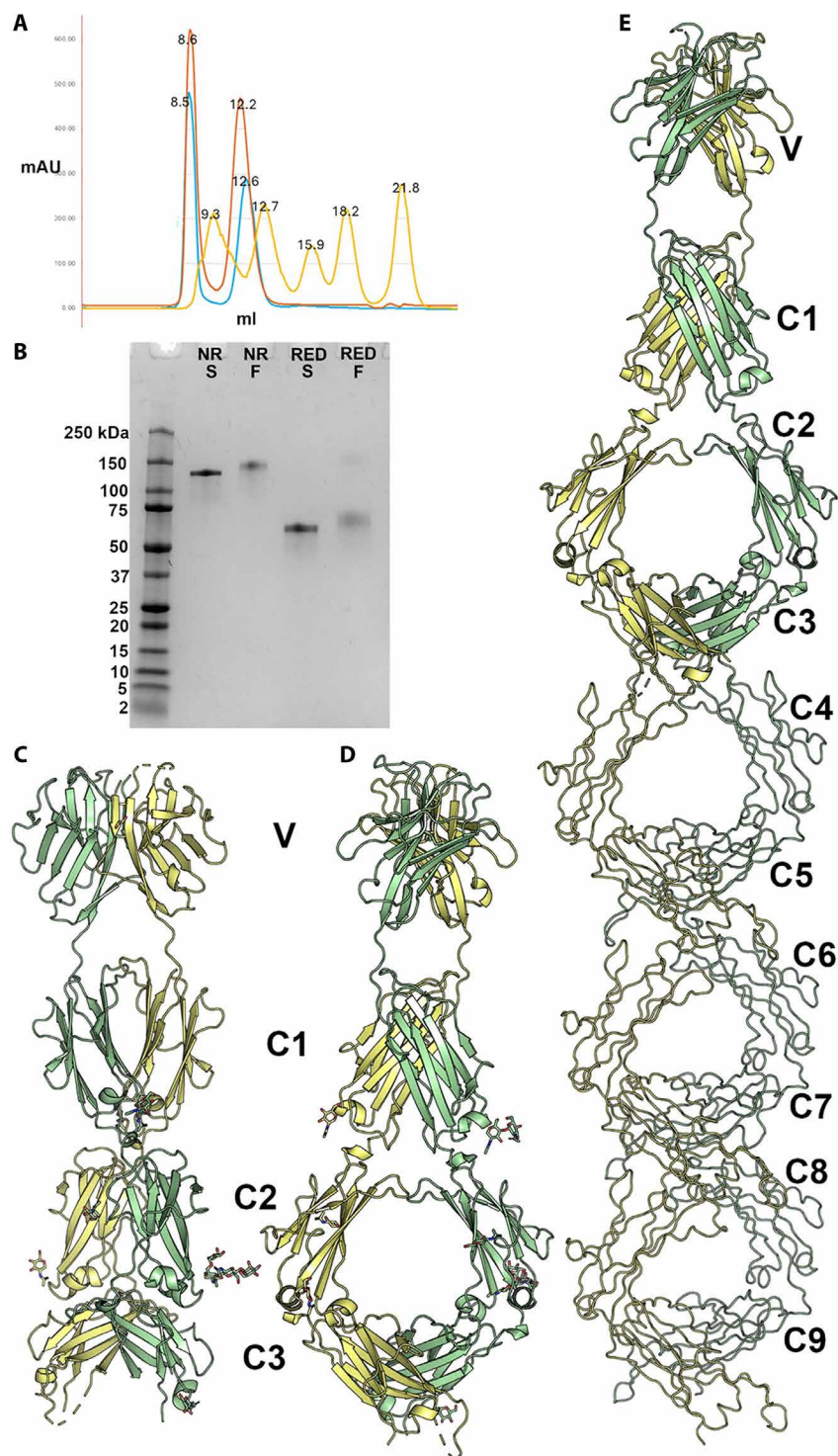


Fig. 6. Utlg2-V-C1-C2-C3 purification and crystal structure. (A) Size exclusion chromatography (S200 10/30, Cytiva) of Utlg2 V-C1-C2-C3. Blue/red = protein produced in Expi293GnTI⁺/Expi293F cells; orange = molecular weight standards (Bio-Rad). The Utlg2 V-C1-C2-C3 elutes at a volume consistent with a homodimer. mAU, milli-absorbance units. (B) SDS polyacrylamide gel (Bio-Rad) of the four N-terminal domains of Utlg2. Starting from the left: Lane 1 = markers (Precision Plus, Bio-Rad); lane 2 = Expi293GnTI⁺ Utlg2-ctag (S), nonreducing (NR); lane 3 = Expi293F Utlg2-ctag (F), NR; lane 4 = Expi293GnTI⁺ Utlg2-ctag (S), reduced (RED); lane 5 = Expi293F Utlg2-ctag (F), RED. (C) The four-domain crystal structure is shown with A and B chains colored light yellow and green. Individual Ig domains are designated V, C1, C2, and C3. (D) Same as (C) but rotated 90° about a vertical axis. (E) Model of the Utlg2 full-length ectodomain. The four N-terminal domains are from the x-ray structure, while domains C4 to C9 (thin tubes) are modeled. If linear as depicted, then the molecule would extend ~335 Å above the membrane.

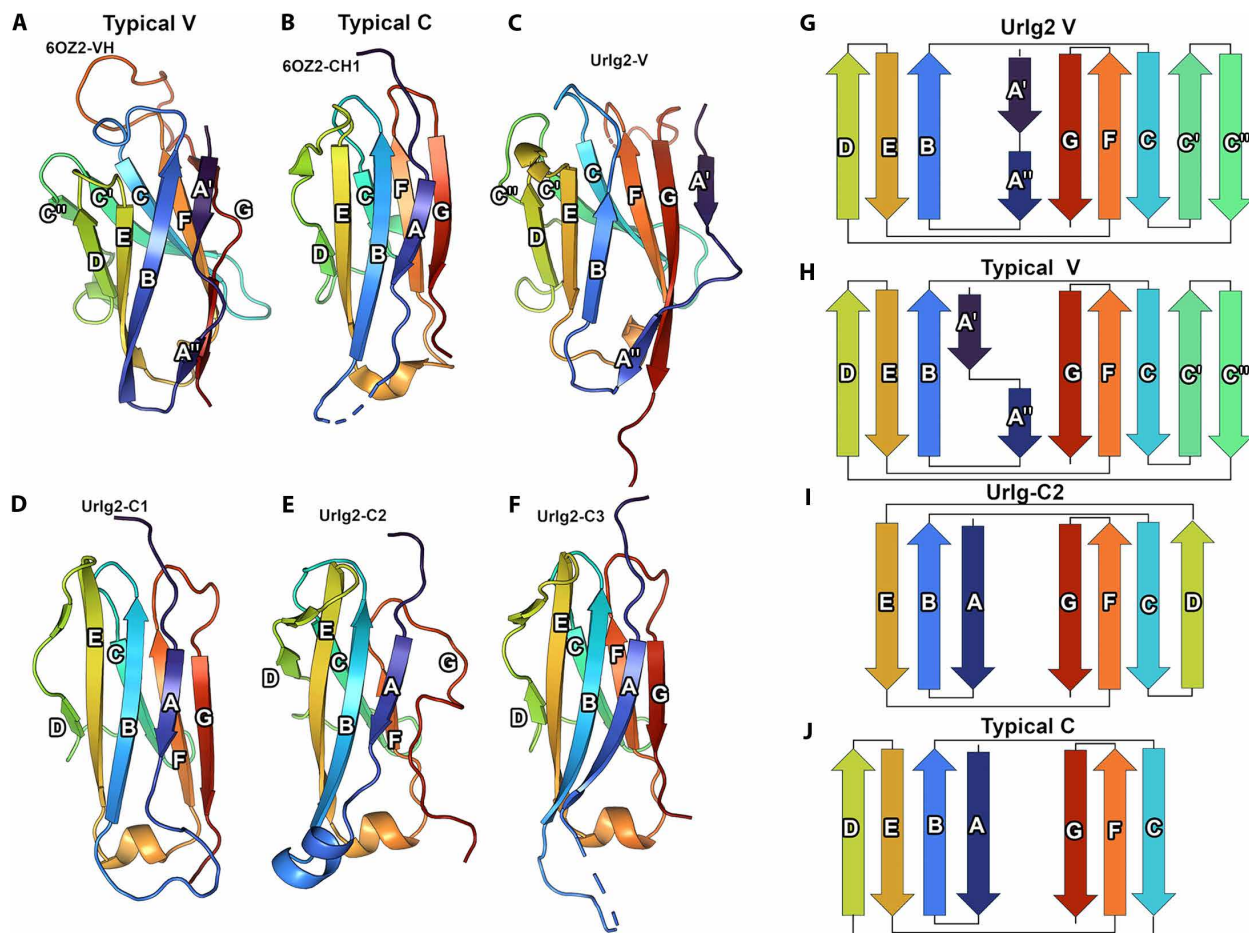


Fig. 7. Urig2 individual Ig domain structure and topology. All domains colored using the Turbo palette with N-C colored from magenta to red. (A) Typical V-type domain from Fab (6OZ2). (B) Typical C-type domain from Fab C_H1 (6OZ2). (C) The Urig2 V domain. (D) Urig2 C1 domain. (E) Urig2 C2 domain. (F) Urig2 C3 domain. (G) Topology diagram for Urig2 V. (H) Topology diagram for typical V-type domain, as in (A). (I) Topology diagram for Urig2 C2 domain, as in (E). (J) Topology diagram for typical C-type domain, as in (B), (D), and (F). The Urig2 V domain lacks a bulge in the middle of its G strand, typically seen in antibody V_H domains. The A' and A'' strands both hydrogen bond to long strand G. The Urig2 C2 domain has an unusual bulge in strand G, and the short D strand hydrogen bonds to strand C rather than to E as in a typical C domain.

them the most promising candidates so far uncovered. It is good fortune that these nonrearranging AgR genes have been retained in sharks (all elasmobranchs so far studied) and no other vertebrates. However, there are other cases in which primordial features of the immune system have been retained in sharks and lost in most other vertebrates, e.g., linkage of the β 2-microglobulin (B2M) gene to the MHC (52), the cluster-type organization of shark IgH genes (18), general conservation of the MHC (17), and primordial RSS sequences (the latter in most fish) (53). In addition, several studies over the past 25 years have shown that cartilaginous fish genomes generally and uniquely among vertebrates evolve at a slower rate than that found in all other vertebrate lineages (12, 25, 54, 55).

Urig2 structure

Nurse shark Urig2 contains 10 IgSF domains, with the last six domains being repeats (>90% sequence identity) of domains C2 and C3 (C2 ~ C4 ~ C6 ~ C8 and C3 ~ C5 ~ C7 ~ C9). The crystal structure of the first four domains (V-C1-C2-C3) reveals a linear, disulfide-bonded homodimeric molecule (Fig. 6) with an unusual V

domain pairing, and C2-C3 pairing highly similar to that found in an Ig Fc fragment, but with no interior interdomain glycan (fig. S4, C and D), unlike Urig (29) that is predicted to have an interior glycan. Modeling of the full-length molecule suggests a V dimer, followed by a C1 dimer, four Fc-like units (Fig. 6E), and, lastly, the TM region. The IgM TM region is highly conserved in vertebrates (56). While the Urig2 TM is only 50% identical to the IgM TM, the highly conserved Leu-Leu-Ile-Phe in the TM center is shared between IgM and Urig2.

While the first four IgSF domains crystallize as a very linear molecule, whether the full-length molecule on the cell is linear, bent, or inherently flexible is not known. The Urig2 IgSF domains all contain the canonical disulfide bond linking the two β sheets within the domain, as well as the conserved Trp residue that flanks the internal disulfide (fig. S4A), and, with some minor variations, are similar to domains from antibodies and TCR molecules (tables S3 and S4).

Structural conservation extends to the presence and absence of the beautifully packed C domain BC-loop cisPro (fig. S4) that is cis or trans depending on domain type: For example, the Pro is trans in

antibody C_H2 domains but cis in C_H1 and C_H3 domains, and the same pattern is seen for the corresponding UrIg2 C2 (trans) and C1/C3 domains (cis) (table S4). Other conserved features include the C domain AB loop that can be found in an extended conformation in antibody C_H1/C_H3 domains but is helical in the C_H2 domain, where that loop closely contacts C_H3. We see the same trend for C1/C3 (extended) and C2 (helical) (Fig. 7). As noted previously (57), it is not uncommon for Ig domain edge strands (A, G, and D) to hydrogen bond to different sheets, and we see slightly unusual edge strand positions for the V and C2 domains of UrIg2 when compared to their closest structural homologs (Fig. 7). The UrIg2 C1 domain is most structurally similar to C_K domains, although it appears in the same position as C_H1 in a modern antibody, directly after V. The C_H1 domain in modern antibodies requires the interchange of BiP for L chain to fold correctly in a quality control scheme to ensure that antibodies have the correct H/L chain pairing (58). Thus, the C_H1-type fold likely developed when L chains were included in the antibody format.

Identification and conservation of the original AgRC and its MHC association in cartilaginous fish

Unexpectedly, all three elasmobranch IgH and some IgL isotype loci are linked to TCR $\alpha\delta$ and *PSMB5* genes (Fig. 1). As discussed above, tight clustering of these genes should allow access to transrearrangements (26) and generation of new genes with chimeric features such as NARTCR (28) and TAILV [TCR δ -associated IgH-like V segment (33)]. In addition, close linkage of V_H and IgNARV genes to the TCR $\alpha\delta$ locus provides a hypothesis as to how both Ig and TCR genes are targeted for somatic hypermutation, unlike almost all other vertebrates that lack TCR mutation (59). Attracting AID activity to this gene complex for affinity maturation of Igs and TCR repertoire shaping also could have facilitated the enzyme's co-option for class-switch recombination in tetrapods (60). Because the potential precursor UrIg2 is a homodimer, and the VJ regions do not associate similarly to typical V_H/V_L (variable region of immunoglobulin light chain) (or TCR $\alpha\beta$ and TCR $\gamma\delta$) in the crystal structure, it is possible that the original AgR was a single-domain VJ, similar to the IgNARV (61). However, unlike IgNARV, the UrIg2V is a dimer interacting via the A and G β strands rather than an entire β sheet (CC'C"FG) found in V_H/V_L. Thus, we speculate that a UrIg2 precursor gave rise to both the NAR-like monomeric V by losing the A/G strand interaction and the dimeric V by gaining intimacy between the domains. The tight genetic linkage also provides the foundation of the translocon IgH locus in all other vertebrates (Fig. 2) (62). Lastly, the genetic connection between Ig and TCR δ further supports the hypothesis that antibodies and TCR $\gamma\delta$ recognize antigens in a similar way, without MHC (or other)-restricting elements (63).

We and others have previously hypothesized that the huchr9, 12, and 14 were derived from the ancient precursor of huchr9 after 2R (9, 10, 16, 17). Mapping of the nonrearranging *UrIg* in the shark MHC (huchr6) was unexpected because it provided direct evidence of Ig (AgR) per se being linked to MHC (29). Here, we show the connection of *UrIg2* to the huchr14/9 MHC paralogous region, further affirming our previous hypothesis that the AgR was derived from the precursor of huchr6/9 at 1R, which could be called the proto-MHC, or now more descriptively, the proto-MHC/AgRC. A hypothetical scenario for the emergence of the different AgRs based on *UrIg*/*UrIg2* being the precursor is shown in Fig. 2. Cartilaginous fish also have a direct connection to huchr9, lost in other vertebrates

(17). Of the four cartilaginous fish L chain isotypes, IgL λ is entirely V-J joined in the germline (32). A cluster of these genes is found in the classical huchr9 MHC paralogue, suggesting (i) a direct connection to the ancient paralogous region and (ii) the split from huchr9 and 14 occurred very early in vertebrate history.

We hypothesize that all jawed vertebrate AgRs were generated in this AgRC. The precursor TCR region duplicated to give rise to TCR $\alpha\delta$ and TCR $\gamma\beta$ loci via cis duplication—this was suggested by Hood and colleagues (64) and is consistent with one TCR gene with D segments and one lacking D, as well as an inverted V segment following the C exon for both the β and δ loci. It would be disadvantageous to have transrearrangements of V_H to TCR β , as those V regions are under different selection for MHC restriction. So, the TCR $\beta\gamma$ complex was translocated, and later, TCR β was separated from TCR γ ; they are on the same chromosome in humans (huchr7) but on opposite sides of the chromosome, possibly via a pericentric inversion. On the other hand, TCR δ V and V_H are the same V type, as suggested by Chien and colleagues (65). As mentioned above, in many nonmammalian vertebrates, TCR δ uses canonical V_H as part of the repertoire, to the point of having V_H genes actually within the TCR δ locus in sharks, frogs, coelacanth, chickens, and others. We are not yet sure how TCR α is shielded from the transrearrangements, but similar to TCR β , they would be disadvantageous. Regulation of the chromatin configuration likely protects TCR α 's purity.

How does the *UrIg* system function?

The *UrIg*/*UrIg2* genes do not function similarly to typical AgR. They are not expressed by lymphocytes but rather at the highest amounts in the liver (both genes) and the gill (*UrIg2*). *UrIg* protein is found in the ECM of several shark tissues (29), suggesting that it may serve an innate protective function, perhaps similar to galectins, carbohydrate-binding protein “multitaskers” that bind to self and foreign ligands in the ECM (66). The gene conversion events found for *UrIg2* domains 4 to 9 could suggest that, similar to protocadherins—IgSF molecules in the brain that regulate neuronal differentiation and undergo gene conversion (67, 68)—*UrIg2* molecules make homophilic interactions between cells. Regardless of their functions, these preliminary expression data suggest a different role for these ancestral molecules compared to AgR, perhaps expected of precursor-related molecules. *UrIg* and *UrIg2* seem to be under quite different selection pressures, with *UrIg* being conserved in cartilaginous fish taxa while *UrIg2* is divergent in sequence and gene structure among species. *UrIg* is encoded in a relatively conserved part of the genome, the class III region of the MHC (17), while *UrIg2* is found in a “malleable” region with the AgR genes that expand and contract rapidly over evolutionary time (31). It will be exciting and challenging to unravel the function of the *UrIg*/*UrIg2* system, whether the molecules function independently or as a team. Regardless of the function, we unveiled the origins of the AgR in the proto-MHC and likely identified the gene or its close relative invaded by the RAG transposon.

MATERIALS AND METHODS

Data mining

IgH chains map to huchr14. We started analyzing the cartilaginous fish IgH chain locus to determine whether this region has homologs to the huchr14 marker genes such as *psmb5*. All gene models for Ig and TCR genes were manually curated using BLASTp and by examining all amino acid sequences for canonical residues. During manual

annotation of Ig genes, novel genes were found from various cartilaginous fish genomes, which are paralogues to the previously found Ig-like gene, *Urlg*. We used V fragments of these new genes as queries for searches against the databases in the National Center for Biotechnology Information (NCBI). Furthermore, paired-end RNA-seq reads from juvenile nurse shark liver (SRX15412459) and gill (SRX15412461) were downloaded from the NCBI Sequence Read Archive (SRA) and de novo assembled using rnaSPAdes, version 3.14.1 (69). All initial hits from the transcriptome were manually annotated to validate the novel genes (e.g., GIWU01221083.1) to obtain the corresponding full-length transcripts. The sequence of the coding region was further confirmed by sequencing the PCR amplicon using nanopore technology, and the assembled sequence was submitted to GenBank (accession number PQ663719). We also searched for full-length sequences in other cartilaginous fish genomic and transcriptomic databases. All species sequences retrieved were partial, except for those from the epaulette shark.

Phylogenetic tree analysis

The V and C types of IgSF domains of the *Urlg2* were separately aligned with other AgR IgSF domains. For the V domain, we removed D elements from the sequence to obtain a better alignment of VJ. Amino acid sequences of V- or C-IgSF exons were aligned using Clustal X multiple sequence alignment, and bootstrapping phylogenetic trees were constructed using MEGAX software (70, 71). We optimized tree construction following MEGAX and chose maximum likelihood and Whelan and Goldman model (71) trees with γ distributions of 2 to 6. The accessions are as follows: *Urlg2*[−] XP_041031358.1 (Caca: great white shark), XP_038644097.1 (Syca: small spotted catshark), GHAY01055342.1 (Etsp: velvet belly lanternshark), HAGT01051777.1 (Sqac: spiny dogfish), XP_048449493.1 (Rhly: whale shark), XP_051865846.1 (Prpe: smalltooth sawfish), XP_060677870.1 (Heoc: epaulette shark), and GJPK01149282.1 (Chgr: gray bamboo shark). Atlantic stingray sequences were reconstructed from genomic assembly of the following gene models: LOC132381413 (Hysa 2.1: Atlantic stingray), LOC132381410 (Hysa), LOC132381409 (Hysa 2.2), *Urlg*-GIWU01140354.1 (Gici), GGGL01499922.1 (Heze), GCC18650.1 (Chpl), and HAGU01063221.1 (Sqac). Other nurse shark genes: AAA50817 (IgM), AAB08972 (IgW), AAB42621 (IgNAR), AAV34678 (IgL), L16765 (IgK), AAV34678 [IgS' (Sp)], ABO64184 (Ig σ), ADW95866, JAHHRZ010000115.1 (TCR α), ADW95903 (TCR β), ADW95942 (TCR δ), ADW95913 (TCR γ), MT-914154-914161 (NCR3 genes), BAK52514.1 (B2M), AAF66110.1 (MHC class I- α), L20275.1 (MHC class IIb), AAA49311.1 (MHC class IIa), and GIWU01258313.1 (class II-W lineage: WAC) (72). Other cartilaginous fish genes: S12838 (Leer IgM), WLO97390 (Okke: ocellate spot skate IgNAR), XP_051883159 (Prpe IgW), AAB51497.1 [Raeg (clearnose skate) TCR δ], AAB51496.1 (Raeg TCR β), AAB51498.1 (Raeg TCR γ), AAB51495.1 (Raeg TCR α), XP_051901911 (Prpe IgLS2), XP_051887747 (Prpe IgLL), XP_051894291 (Prpe IgLS), XP_060707137.1 (Heoc TAPBP), the germline-joined spotted ratfish *Hydrolagus* IgM V (Hycy-gjH), single-domain *Hydrolagus* IgM V (Hycy-SD), and the conventional IgM (AAC-12891) (73, 74).

RNA sequencing

Total RNA from 20 tissues [brain: diencephalon, cerebellum, medulla, midbrain, and cerebrum; colon; epigonal organ; gill; kidney; liver; muscle; olfactory epithelium; pancreas; rectal gland; spiral valve;

spleen; stomach; thymus; white blood cells; and uterus (submitted to SRA under accession number PRJNA1179899)] was used in transcriptome sequencing (bulk RNA-seq) to assess the levels and distribution of gene expression of the *Urlg2* gene. Tissue-specific mRNA libraries were obtained using poly(A) enrichment and sequenced on a NovaSeq S4 platform aiming at 20 million paired-end reads per tissue (150PE). Raw read quality was checked using the FastQC software (www.bioinformatics.babraham.ac.uk/projects/fastqc; v0.11.9), and adapter removal for Illumina Universal Adapters was performed using Trimmomatic (v0.39) (75). Gene expression quantification was estimated with RSEM (76) based on read mapping performed with Bowtie2 (77) using default parameters for pair-end reads ("--sensitive --dpad 0 --gbar 99999999 --mp 1,1 --np 1 --score-min L,0,-0.1 --no-mixed --no-discordant") against the full-length *Urlg2* sequence. The gene expression levels among tissues for the *Urlg2* gene are in TPM.

Reverse transcription polymerase chain reaction

The first-strand cDNA was synthesized using SuperScript IV First-Strand Synthesis System (Invitrogen) from 500 ng of total RNA of nurse shark "NJ." After synthesis, the cDNA was diluted with 50 μ l of EGTA, and 1 μ l of the diluted cDNA solution was used for PCR amplification with the GoTaq Master Mix (Promega). PCR conditions are as follows: 95°C for 2 min, followed by 35 cycles of 95°C for 45 s, 60°C for 45 s, and 72°C for 1 min with final extension at 72°C for 5 min. The interexon primers used for the reactions are V forward 5'-tgcccagcgaatggagagtg-3' and C1 reverse 5'-gaactccgattctcatccac-3', and the expected amplicon size was 238 base pairs. We used nucleoside-diphosphate kinase (NDPK) as a PCR control, although NDPK expression is known to be lower in some tissues, such as muscle and pancreas.

NbSyn2 and *Urlg2* protein expression and purification

A gene encoding a C-terminally His-tagged nanobody NbSyn2 that recognizes the EPEA C-tag (78) was synthesized (Integrated DNA Technologies) and cloned into vector pET26b. Transformed BL21 Star DE3 cells (Invitrogen) were grown in Luria-Bertani broth (500 ml in 2 liters of baffled flasks; shake speed = 100 rpm) at 37°C until an optical density (OD) of ~0.7, followed by induction with 1.5 mM isopropyl- β -D-1-thiogalactopyranoside. Flasks were then shaken for ~16 hours at 28°C, and the resulting cells were harvested by centrifugation at 4000 $\times$ g for 20 min, disrupted 3 $\times$ at 15,000 psi (103.4 MPa; Avestin), and clarified by centrifugation at 6000 $\times$ g for 45 min. Filtration of the lysate was performed with a 0.22- μ m filter (MilliporeSigma). Nanobody was affinity purified from lysate using Ni-agarose resin and then further purified by size exclusion chromatography with a Superdex75 16/60 column (Cytiva) with running buffer 20 mM Tris and 150 mM NaCl (pH 8). Purified nanobody was coupled to activated CNBr Sepharose 4B (Cytiva) following the manufacturer's protocol.

Urlg2 [domains V-C1-C2-C3 with a C-terminal EPEA (C-tag)] (79) and the last five residues of the shark leader at the N terminus were cloned into a phCMV3 vector with an Igk leader and expressed in either expi293F or expi293 [lack of N-acetylglucosaminyltransferase I (GnTI)] cells (Thermo Fisher Scientific) following the manufacturer's standard protocol [see extended data Fig. 4 for the sequence of the crystallized construct]. The expi293(GnTI) cells lack the GnTI enzyme and thus do not make protein with complex N-glycans as can be found on protein expressed in expi293F cells. Protein was affinity purified from clarified media using NbSyn2/C-tag affinity

resin and eluted with 2 M MgCl₂ and 20 mM tris (pH 7.6), followed by size exclusion chromatography on a Superdex200 16/60 column with running buffer 20 mM tris (pH 8) and 150 mM NaCl. Proteins were concentrated to ~10 mg/ml and screened for crystallization using our Rigaku CrystalMation system at Scripps Research with screens JCSG 1-4 at 4° and 22°C. The crystal used for data collection was obtained from the protein made in expi293F cells and precipitant 0.2 M magnesium formate, 20% polyethylene glycol, molecular weight 3350, and 0.1 M MES (pH 7). The crystal was cryoprotected with the well solution augmented with 20% ethylene glycol and flash cooled by rapid plunging into liquid nitrogen.

X-ray data collection and structure determination

Data were collected from a single crystal at 100 K using the Advanced Light Source beamline 5.0.2 with a wavelength of 1.0000 Å. Data were processed with HKL-2000 (80). The crystal structure was phased by molecular replacement (MR) using Phaser (81). Models in the Protein Data Bank (PDB) with about 40% sequence identity were available for the C1, C2, and C3 domains, while sequence identity for V domain models was lower, with the highest sequence identity in the PDB at about 30%. The asymmetric unit was estimated to contain two V-C1-C2-C3 chains (eight Ig domains), based on a Matthews coefficient (82) of 2.46 Å³/Da (~50% solvent) for a dimer and a noncrystallographic twofold peak in the self-rotation function at $\omega = 90^\circ$, $\phi = 46^\circ$ and $\kappa = 180^\circ$. MR was first attempted with models generated by AlphaFold2 (83, 84) for individual Ig domains and Ig domain dimers and the V-C1 dimer and the C2-C3 dimer; however, no convincing solutions were obtained using these models. Extensive searches using Ig domains extracted from Fab and Fc structures in the PDB were then implemented using Phaser within a Unix script to iteratively evaluate many different models. The final models used for the MR solution were the C_L domain from 1IGJ (UrIg2 C1 domain), shark IgNAR C3 domain from 4Q9C (UrIg2 C2 domain), C_L domain from 1PG7 (UrIg2 C3 domain), and V_L domain from 6WX1 (UrIg2 V domain). Structure refinement was conducted with Phenix (85) and model adjustments with Coot (86). The final model has 95.11, 4.54, and 0.35% of residues in Ramachandran favored, allowed, and outlier regions and an all-atom clashscore of 5.7 as determined by MolProbity (87). Most residues in A and B chains were modeled (A1-103, A110-351, A359-447, B0-350, and B364-447) (see extended data in Fig. 4 for sequence and amino acid positions). Partial glycans were modeled at positions A144, A253, A328, B144, B253, and B328, with density at B253 for a fucosylated NAG-NAG-BMA. Final data collection and refinement statistics are listed in table S2. Buried surface areas were calculated with MS (88), and structure figures were generated with PyMol (<https://pymol.org/>). The Dali server (table S3) (46) was used to search for structural homology for individual domains for individual V, C1, C2, and C3 domains, and comparisons of individual domains were also implemented with CCP4 Superpose (table S4) (89).

Supplementary Materials

The PDF file includes:

Figs. S1 to S5
Legend for table S1
Tables S2 to S4

Other Supplementary Material for this manuscript includes the following:

Table S1

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